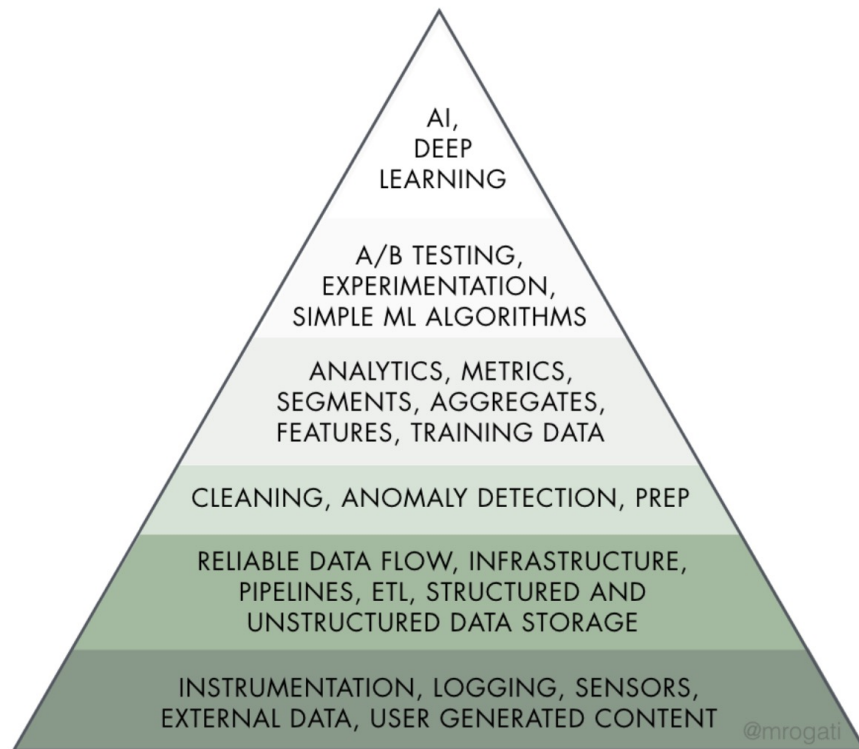


Data Analytics & Data Engineering



Data Integration

Damrongrit Setsirichok

Waranyu Wongseree

หัวข้อ

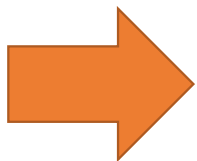
- Data Enrichment
- Data Integration
- Data Pipeline
- Use Case

ปัญหาหลักในการเก็บข้อมูลและเชื่อมโยงข้อมูล

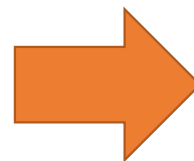
- อยากวิเคราะห์ข้อมูลแต่ไม่รู้จะเก็บข้อมูลอย่างไร
- มีข้อมูลแต่ไม่รู้จะใช้อย่างไร



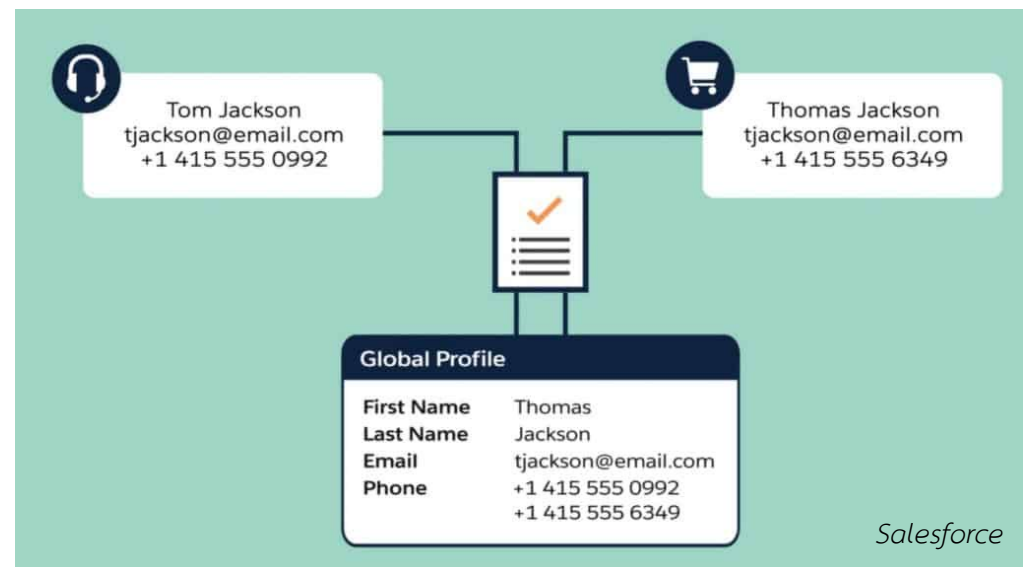
Data Enrichment ทำข้อมูลที่มีอยู่ให้สมบูรณ์ขึ้น



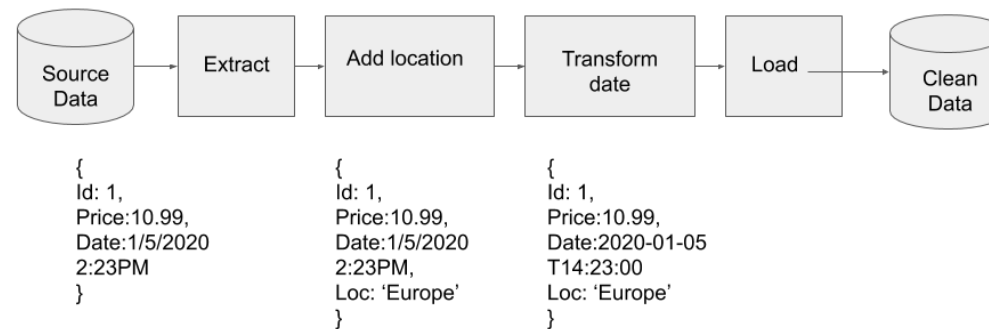
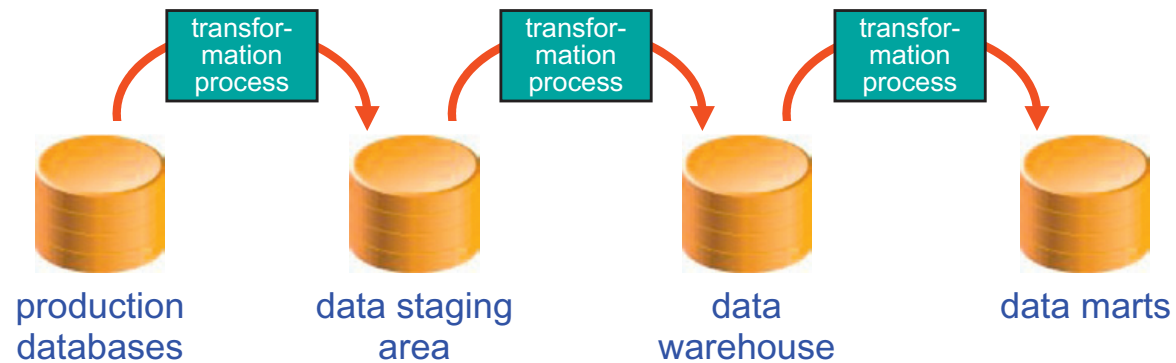
ออกแบบ Data Integration



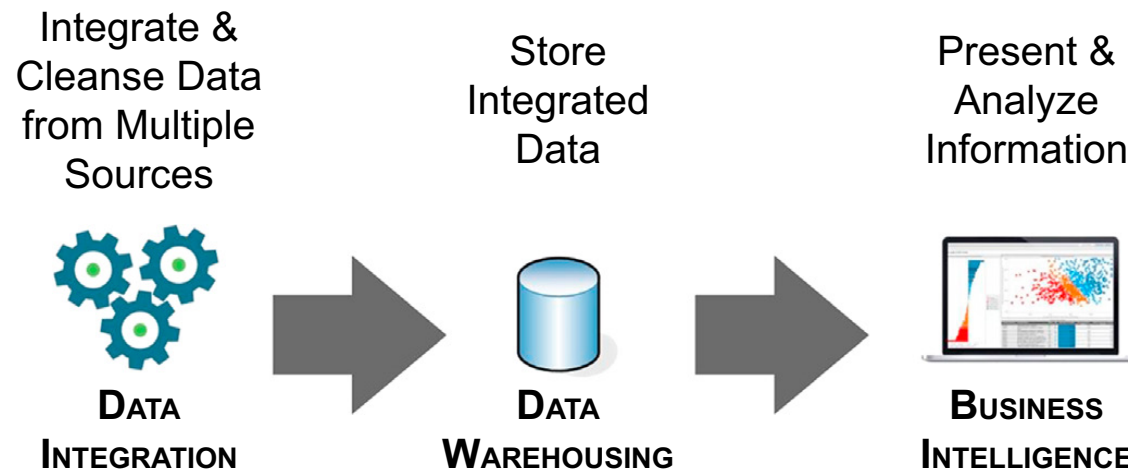
ข้อมูล 360 องศา



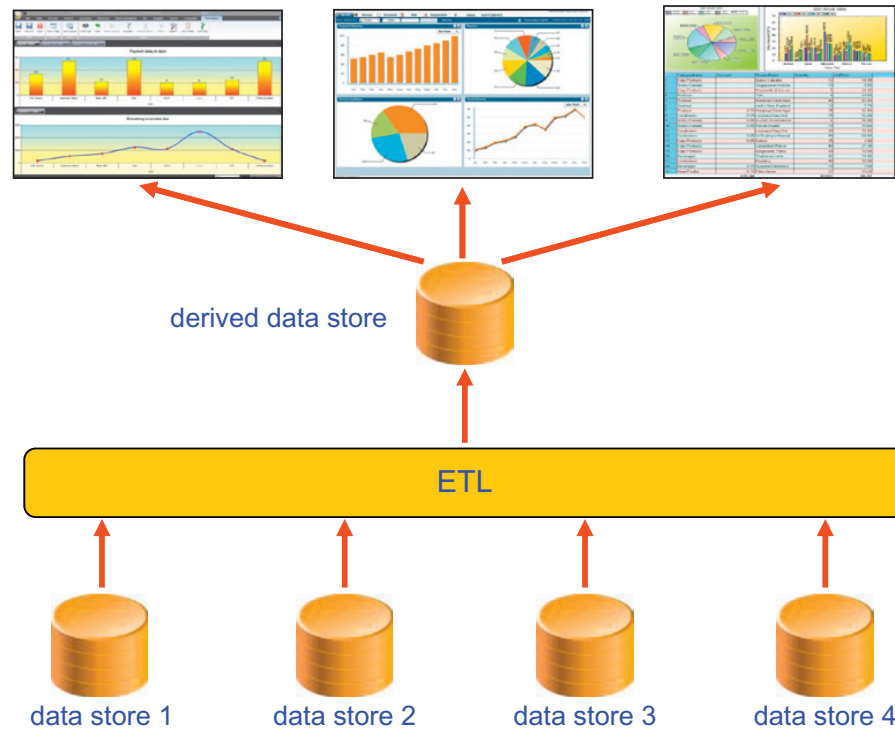
A Business Intelligence system



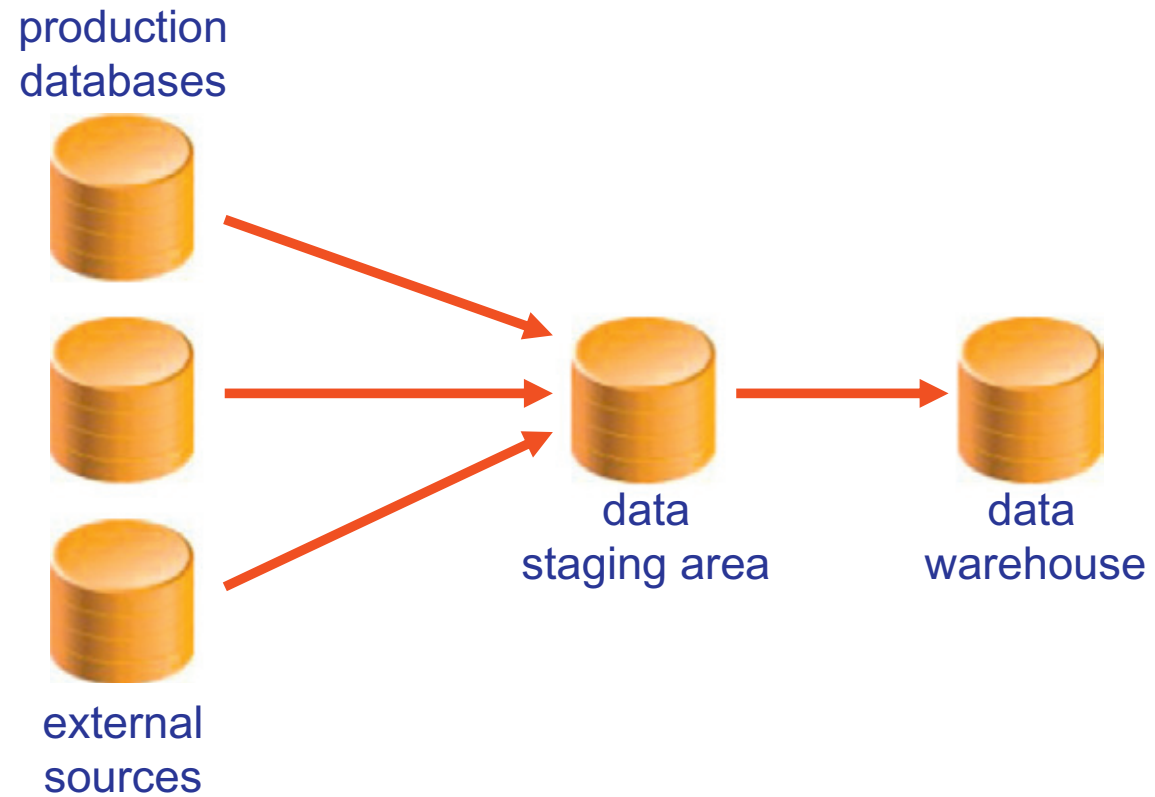
The Business Demand for Data, Information, and Analytics



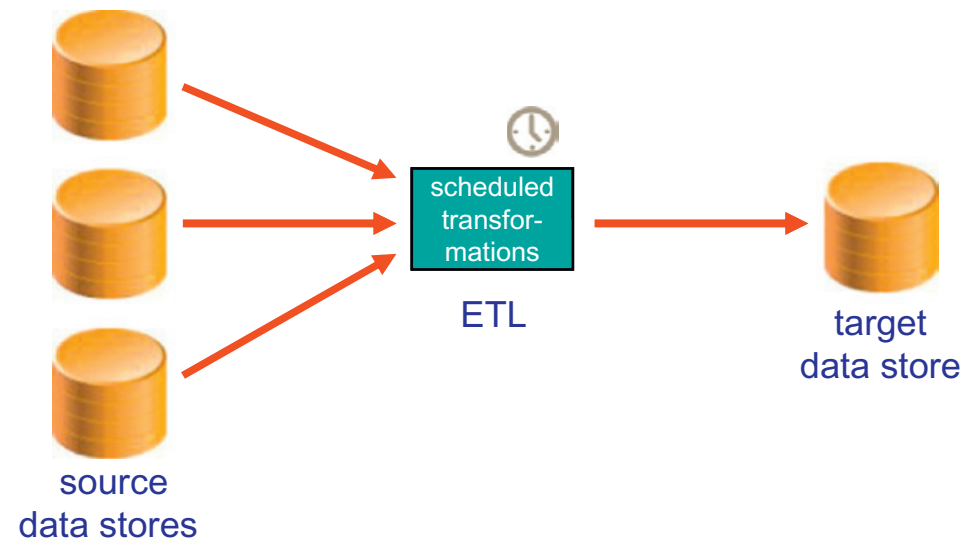
Data Integration ETL



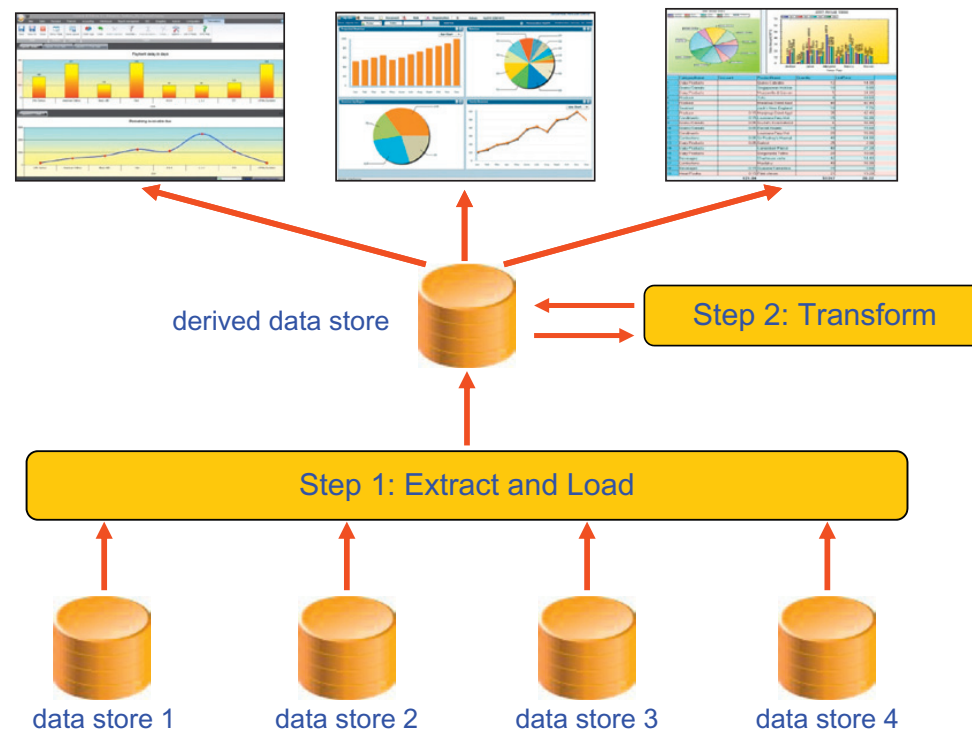
Data staging area



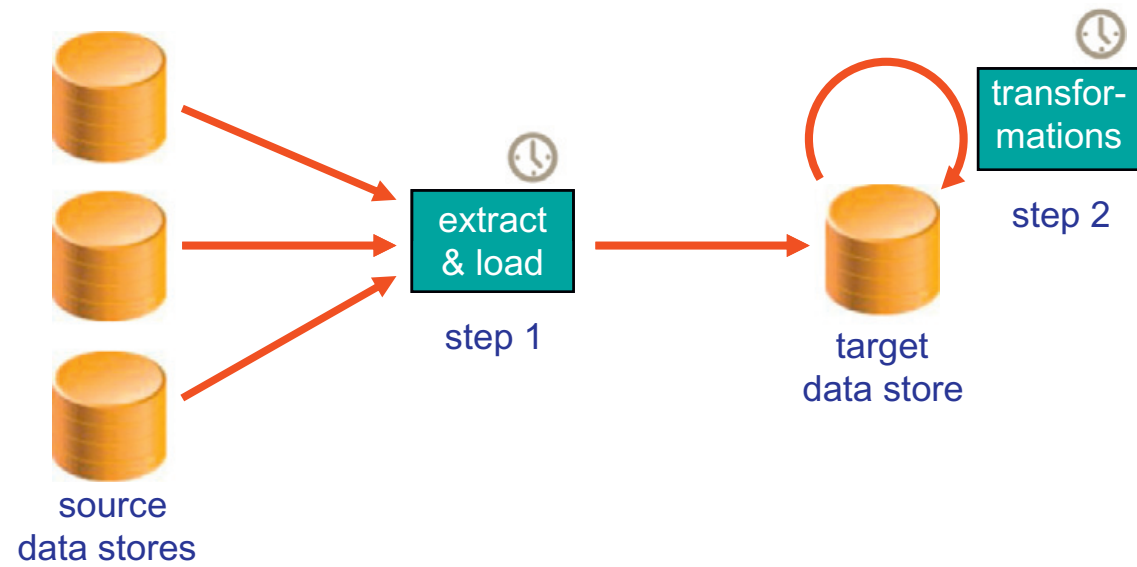
ETL



Data Integration ELT



ELT



ELT vs ETL

ELT

- ดำเนินการที่ปลายทางการเก็บข้อมูล
- ยึดหยุ่นกับผู้ใช้
- รองรับการนำข้อมูลขึ้นวิเคราะห์

ETL

- ETL (Extract, Transform and Load)
- ดำเนินการระหว่าง data pipeline
- กระบวนการที่ใช้เวลามากอาจส่งผลกระทบต่อประสิทธิภาพโดยรวม

ELT vs ETL

Data Lake

- เน้นที่เก็บข้อมูลดิบ (Raw Data) รักษาความครบถ้วนก่อนแปลงข้อมูลและประมวลผล
- เน้นหลักการทำ ELT
- Schema on Read

Data Warehouse

- เน้น ETL (Extract, Transform and Load)
- Schema on Write

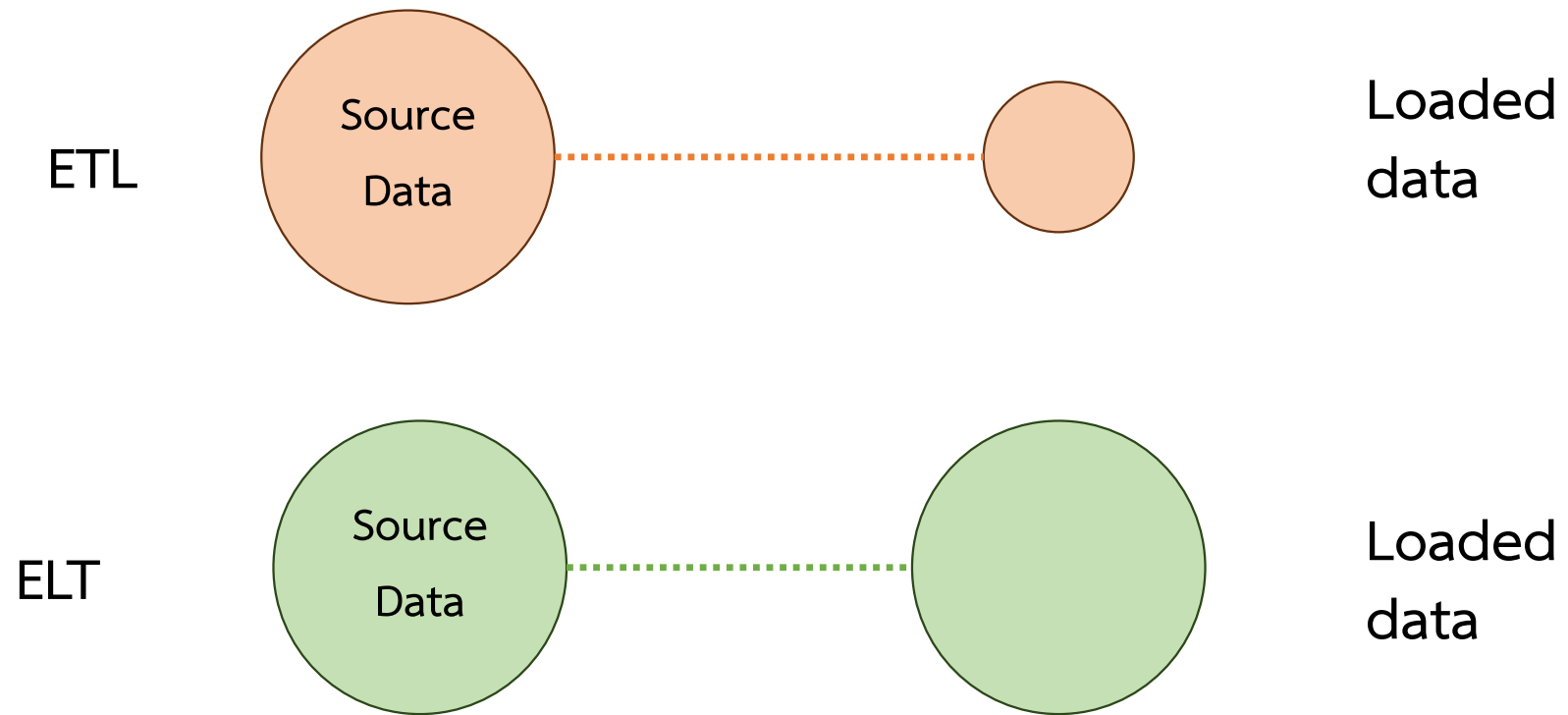
Data Extraction Techniques

- APIs
- Database querying
- Edge computing
- Biomedical devices
- OCR
- ADC sampling, CCD sampling
- Mail, phone or surveys
- Web scraping

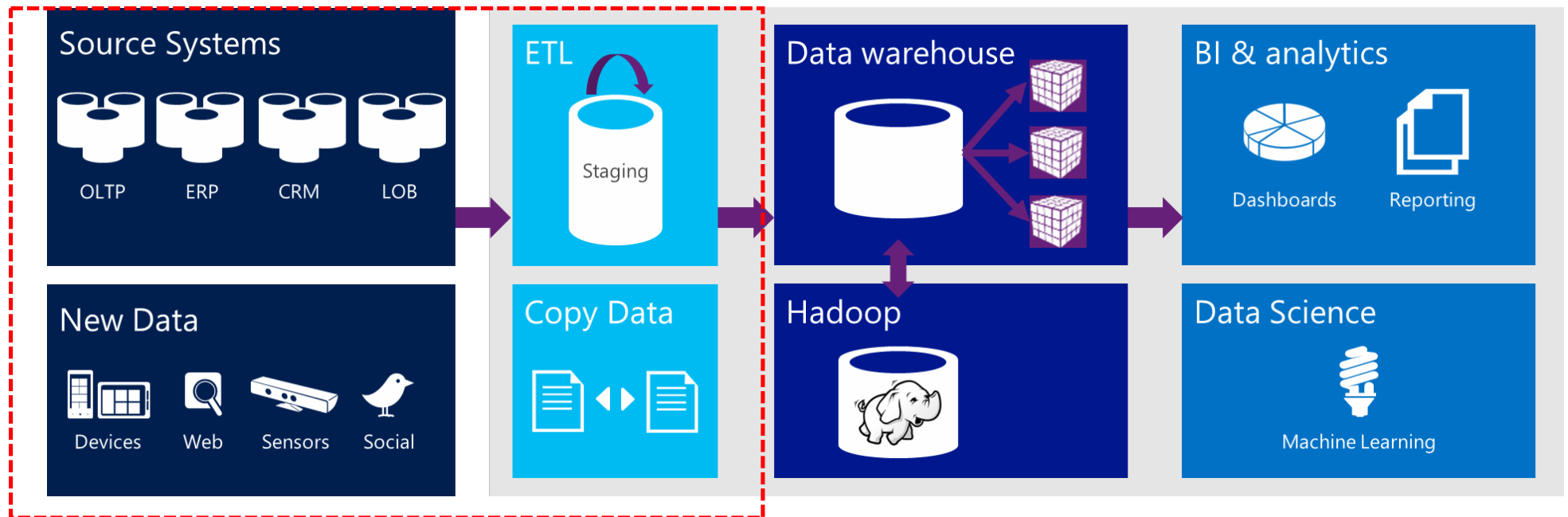
Data Transformation Techniques

- Data Typing
- Data structuring
- Anonymizing, encrypting
- **Cleaning**: duplicate, missing values
- **Filtering**, sorting, aggregating, binning
- **Normalizing**: converting data to common units
- **Joining** data source

Information loss in transformation



Data Load Techniques



Multiple data streams

Data Load Techniques

Push-Pull

- Server pushes data to clients: notification, msg
- Client pull requests: RSS feeds, email

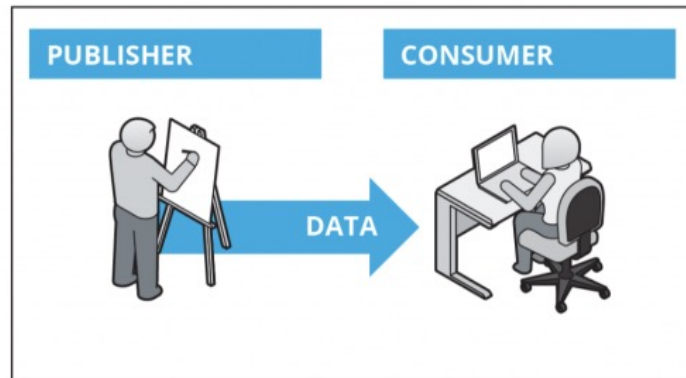
Batch and stream

- Periodic batch update
- Continuous update as data stream arrives
- Micro-batch: smaller and processed more often

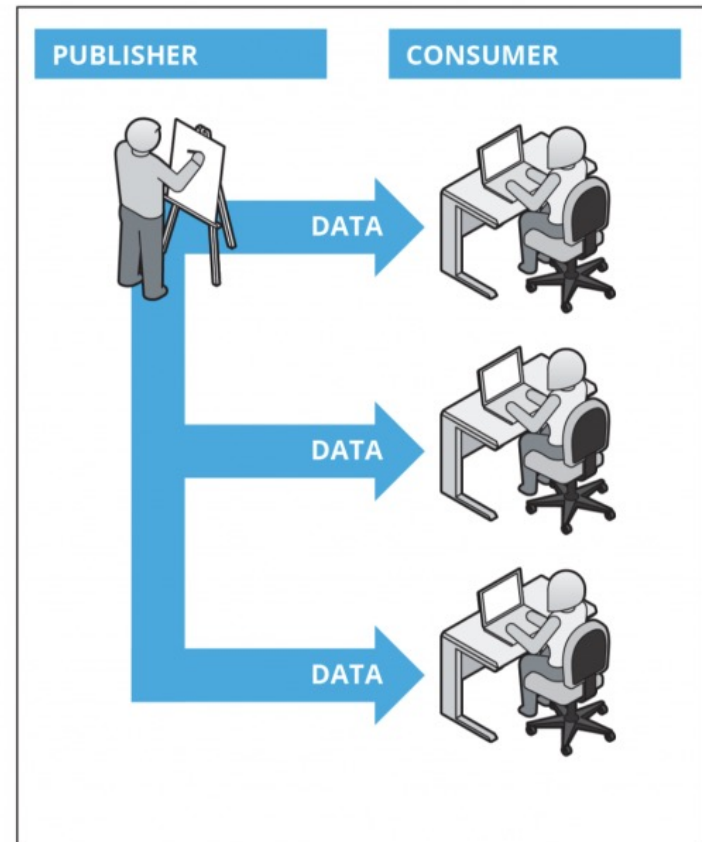
Data Integration Flow

- Raw
- Clean
- Formatted

EASY



CHALLENGING

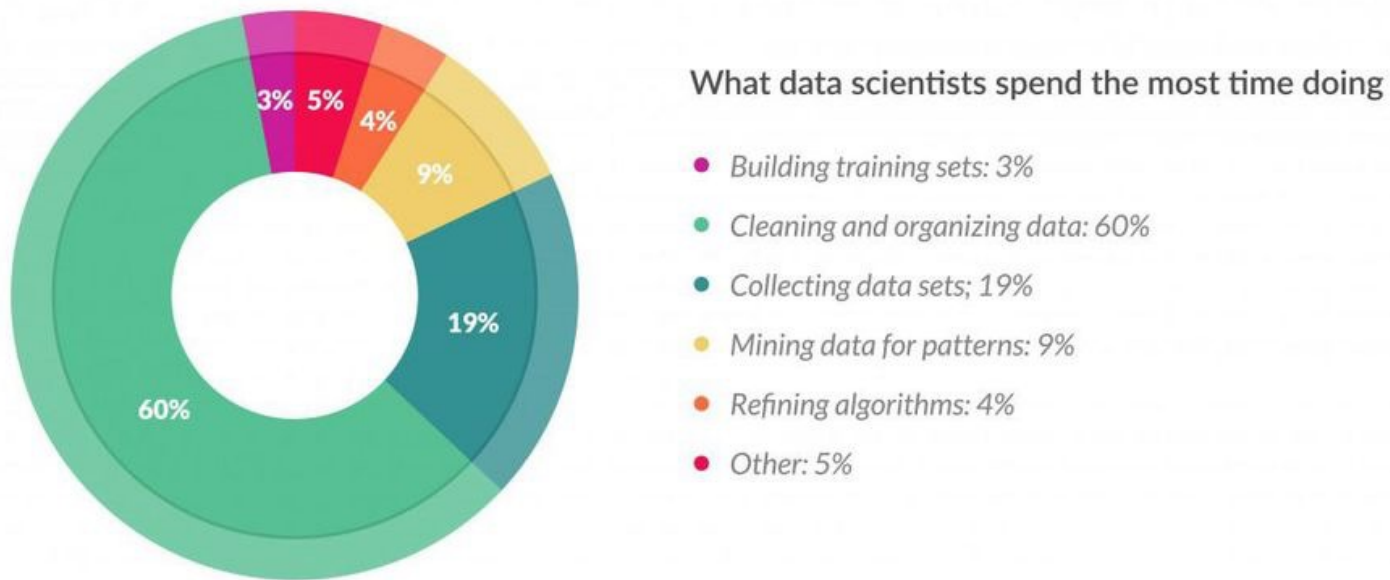


Data Integration Tools

- เครื่องมืออัตโนมัติช่วยการรวบรวมข้อมูลให้มีประสิทธิภาพ
- จำเป็นต่องาน Data Engineer
- Data Delivery จัดการข้อมูลเข้าสู่การเตรียมและส่งออกข้อมูล
 - filter
 - standardization
- Data Accuracy บริหารจัดการคุณภาพข้อมูล
 - duplication
 - Missing value
- Workflows as DAGs



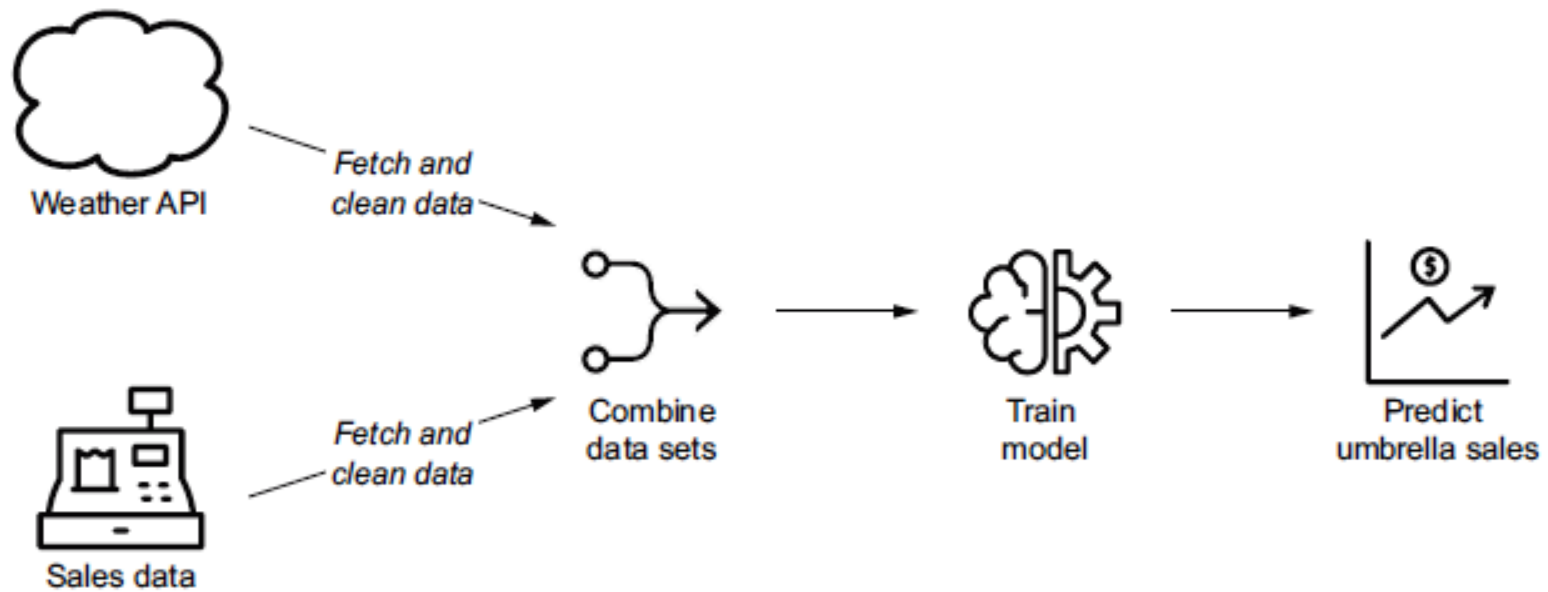
Data preparation accounts for about 80% of the work of data scientists



<https://www.forbes.com/sites/gilpress/2016/03/23/data-preparation-most-time-consuming-least-enjoyable-data-science-task-survey-says/?sh=5d208d216f63>

Data Pipeline

- Series of connected processes



Data Pipeline

Batch pipeline use case

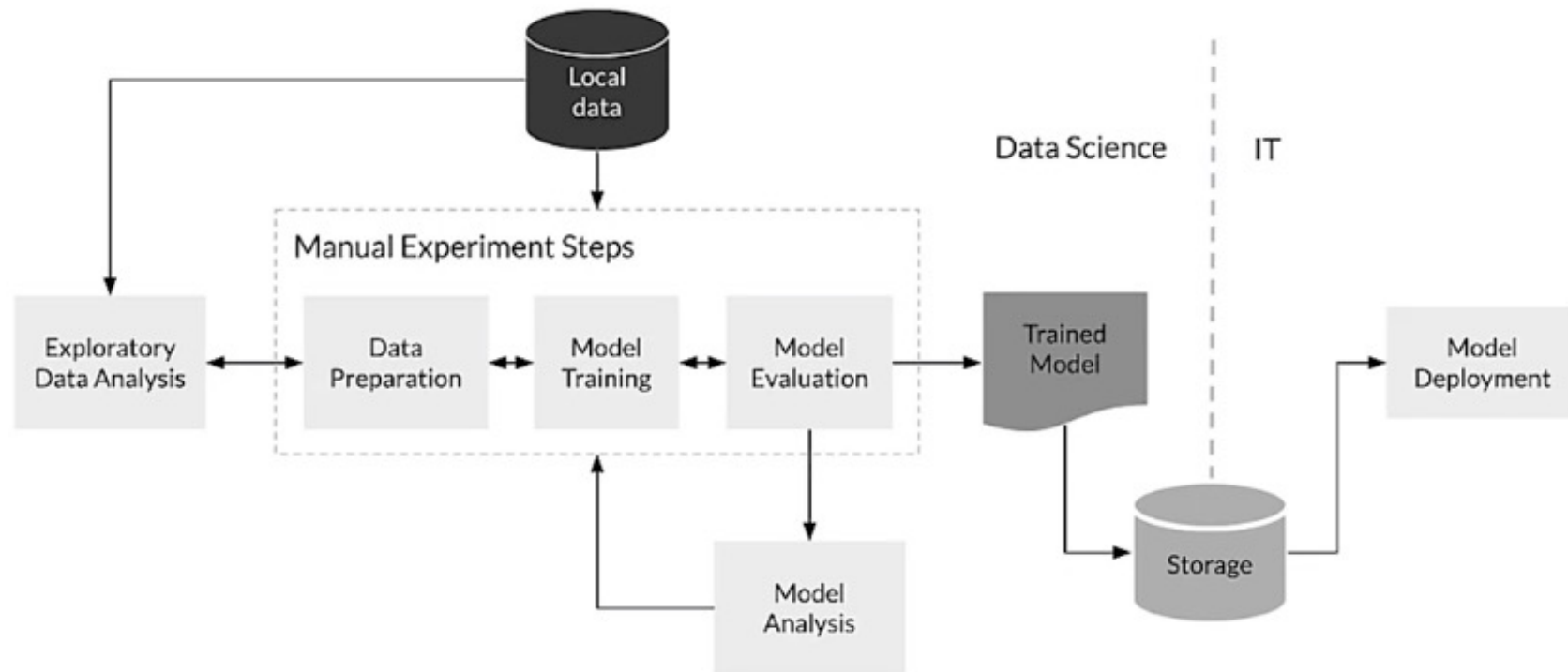
- Data backups
- Transaction history loading
- Billing and order processing
- Forecasting sales or weather
- Diagnostic medical image processing
- Data modelling

Streaming pipeline

- Movies, music or podcasts
- Social media, sentiment analysis
- Fraud detection
- User behavior, advertising
- Stock market trading
- Real-time process
- Recommender systems

MLOps level 0: Manual process

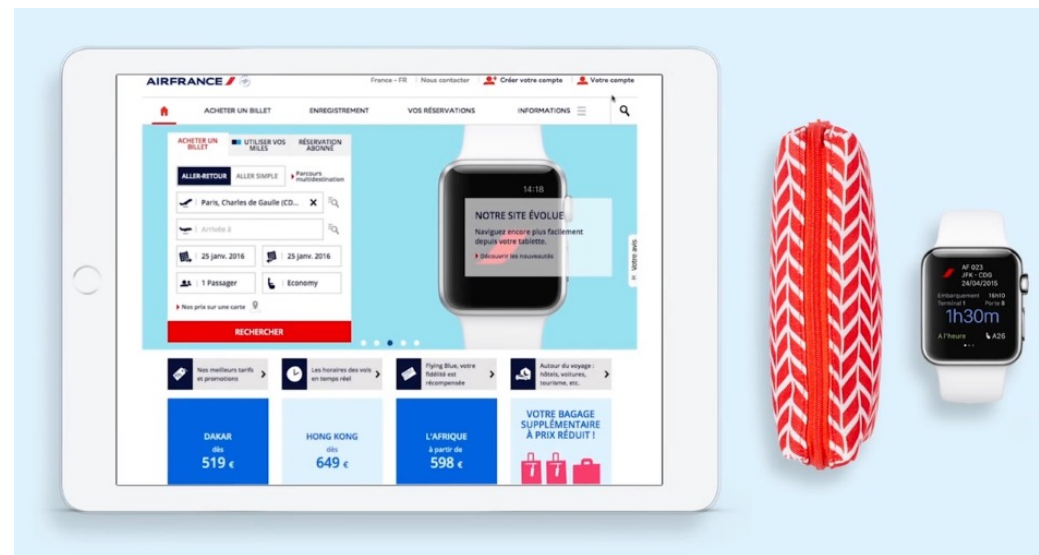
Less frequent releases, so no CI/CD





the 360-degree customer view

- 1 million corrected data entries / month
- 1 billion emails/year





“easy handle big data and generate Java code”

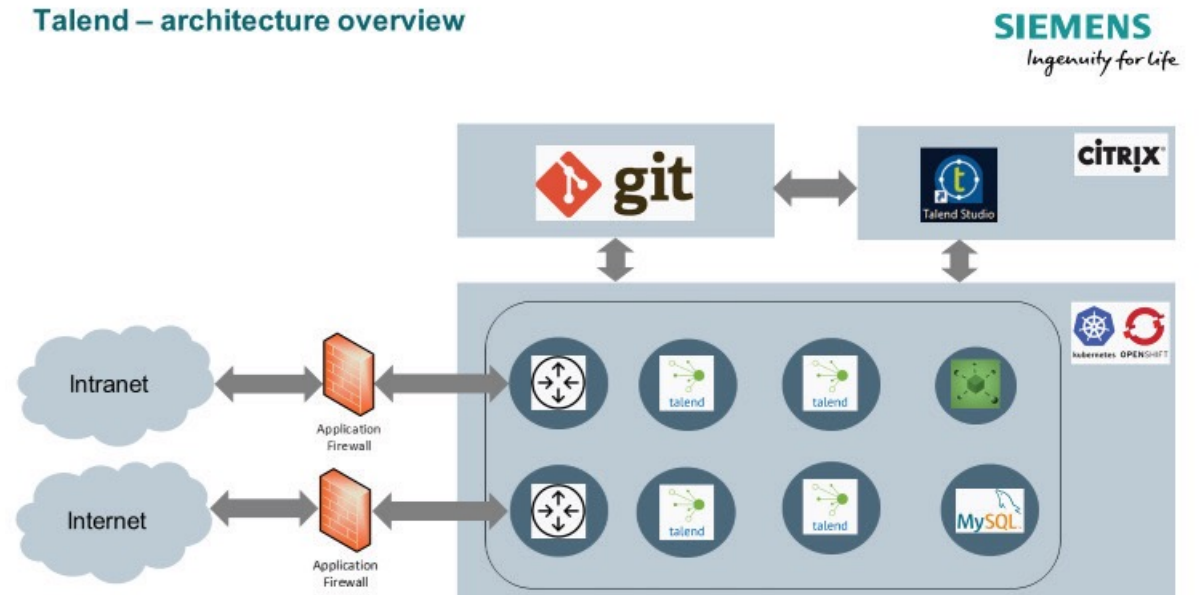
INDUSTRY

- Automotive

INFORMATION

- HQ: Japan
- 10,001+ employees

Talend – architecture overview



Frei verwendbar © Siemens 2019
Page 1

Siemens IT – innovative. reliable. collaborative.

“we could capture data at high speed from hundreds of data sources.”



“Enabled 90 percent of data to be ready for analysis within 3 minutes”

“Enabled consumer segmentation for more personalized marketing”

ALDO

The Lenovo logo, consisting of the word "Lenovo" in white bold sans-serif font on a red rectangular background.

Lenovo

- 18% increase in attach rate for ThinkPad laptop series
- 11% increase in revenue per retail unit with conjoint analysis
- \$1M reduced operating cost within 6 months



- Demos created in 3 days
- Increased productivity
- 150+ users work independently with data



- Thousands of business users working with data
- 30 hours saved per quarter with process automation



Q&A